Software Engineering Assignment

A Project Report On
"Online Learning Management System"

A Project Report Submitted

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INTRODUCTION

In today's fast-paced world, where convenience and flexibility are highly valued, online learning systems have emerged as the top choice for training and education. Throughout the years, these systems have been steadily gaining popularity as a method for teaching new knowledge and skills.
And with the continuous growth of technology, online learning has become more accessible, offering a wide range of benefits to learners of all backgrounds. Gone are the days when learners had to adhere to strict schedules or physically attend classes.

With online learning, individuals can access educational materials and engage in interactive learning experiences from the comfort of their own homes, at their own pace, and at a time that suits them best.

PROBLEM STATEMENT

In today's rapidly evolving educational landscape, traditional one-size-fits-all approaches to learning are proving to be insufficient in meeting the diverse needs and learning paces of students. The advent of online learning systems has opened up new opportunities for remote education, but these platforms often lack the ability to adapt to individual learning styles, preferences, and progress. The need for a more personalized and effective online learning experience has never been greater.

My proposed project aims to address these challenges by developing an online learning system. The system will harness to create a dynamic, adaptable, and engaging learning environment for students.

FEASIBILITY STUDY

- **TECHNICAL FEASIBILITY: -**

The technical feasibility of an Online Learning Management System encompasses various elements:

i. **Infrastructure**: - Assessing hardware and software requirements is crucial. Scalability, compatibility, and performance of servers, databases, and networks are pivotal considerations. Ensuring the system is adaptable to different devices and browsers is essential for user accessibility.

ii. **Development Tools and Technologies**: - Choosing appropriate programming languages, frameworks, and development methodologies is vital. Leveraging technologies for user-friendly interfaces, interactive features, and data management (like cloud storage) enhances the system's efficacy.

iii. **Security Measures**: Implementing robust security protocols to protect user data, prevent unauthorized access, and ensure secure transactions is imperative. Encryption, authentication mechanisms, and regular security updates are fundamental aspects.

iv. **Integrations**: Compatibility with third-party tools and APIs for functionalities like payment gateways, content management systems, and analytics tools should be considered for seamless operations.

- **ECONOMIC FEASIBILITY: -**

Evaluating the economic viability involves assessing the costs and benefits associated with the online learning management system project:

i. **Cost Analysis**: - Estimating development costs, infrastructure expenses, licensing fees, maintenance, and potential future upgrades is crucial. Balancing these expenses against the anticipated returns or cost savings is necessary.

ii. **ROI Projection**: Forecasting the return on investment (ROI) by considering factors like increased user base, potential subscription or usage fees, reduced administrative costs, and improved efficiency through automation.

iii. **Risk Assessment**: Identifying potential risks such as market competition, technological advancements, and changing user preferences helps in devising risk mitigation strategies.

iv. **Business Sustainability**: Ensuring the Learning Management System can generate revenue, sustain operations, and adapt to market changes is essential for long-term economic feasibility.

- **OPERATIONAL FEASIBILITY**

Operational feasibility focuses on the practicality and effectiveness of implementing and running the Learning Management System:

i. **User Acceptance:** Understanding users' needs, preferences, and behaviours through surveys, feedback, and usability testing ensures the system aligns with their expectations, enhancing adoption.

ii. **Training and Support**: Providing adequate training resources and support for administrators, instructors, and users is vital for seamless onboarding and sustained usage.

iii. **Scalability and Maintenance**: Ensuring the system can accommodate growth in user base and content while maintaining performance levels. Regular updates, bug fixes, and support services are crucial for smooth operations.

iv. **Change Management**: Implementing strategies to manage organizational changes resulting from the introduction of the LMS, including communication plans, stakeholder involvement, and transition processes, is essential.

SRS DOCUMENT

1. Introduction

i) Purpose: The purpose of this Software Requirements Specification (SRS) document is to outline the functional and non-functional requirements of the "Online Learning Management System." This document serves as a reference for developers, designers, testers, and other stakeholders involved in the project.

ii) Scope: The Online Learning Management System is designed to provide learners with a personalized and adaptive online learning experience. The platform analyse user behaviour, preferences, and learning styles to deliver customized content, recommendation and support.

iii)Document Conventions UI:
User Interface API: Application
Programming Interface DB: Database

2. Overall Description

i) Product Perspective: The Online Learning Management System is a standalone system that interacts with users through web and mobile interfaces.

ii) Product Features:

Key features of the platform include:

- User registration and authentication
- User profile management
- Content management
- recommendation engine
- Delivery of personalized learning materials
- Progress tracking and reporting
- Communication and collaboration tools

- Payment processing
- Admin panel for platform management

iii) User Classes and Characteristics: The platform will cater to the following user classes:

- Learners: Individuals seeking online education.
- Instructors: Content creators and educators.
- Administrators: Platform administrators responsible for system management.

iv) Operating Environment: The platform will operate in a web and mobile environment. It should support the latest versions of major web browsers (Chrome, Firefox, Safari, Edge) and mobile operating systems (iOS and Android).

v) Design and Implementation Constraints:

The platform must adhere to design and implementation constraints, including security standards, scalability requirements, and third-party service integration.

vi) Assumptions and Dependencies: Internet connectivity is assumed for all users. The platform may depend on external services and content providers.

3. Functional Requirements

3.1. User Registration and Authentication

i) *User Registration*

- Users can create accounts by providing essential information.
- Users can sign up using email/G-Mail.

ii) *User Authentication*

- Users must log in to access the platform.
- Password reset and account recovery options will be available.

3.2. User Profile Management

i) *Profile Creation*:

• Users can create and edit their profiles.

• Profile information includes name, profile picture, contact information, and bio.

ii) *Preferences*:

• Users can set learning preferences, such as subjects of interest and preferred learning styles.

3.3. Content Management

i) *Content Upload:*

• Instructors can upload various types of learning content (Videos, articles, quizzes, etc.).

• Content should support multimedia formats.

ii) Content Categorization

• Content should be categorized by subject, difficulty level, and type.

3.4. Recommendation Engine

i) *Personalized Recommendations:*

• The System should analyse user behaviour and preferences to provide personalized content recommendations.

ii) *Adaptive Learning Paths:*

• The platform should suggest customized learning paths based on the user's progress and goals.

3.5. Learning Material Delivery

i) *Content Access:*

• Users should be able to access recommended content.

• Content delivery should support both streaming and downloadable formats.

3.6. Progress Tracking and Reporting

i) *Progress Monitoring:*

• Users can track their learning progress, including completed courses and achievements.

ii) *Reporting:*

• Generate detailed progress reports for learners and instructors.

3.7. Communication and Collaboration Tools

i) *Messaging:*

• Users can communicate with instructors and other learners.

ii) *Discussion Forums:*

• Provide discussion forums for course-related discussions.

3.8. Payment Processing

i) *Subscription Plans:*

• Offer various subscription plans for premium content.

• Support one-time purchases for individual courses.

• Giving a month free trial

ii) *Payment Gateways:*

• Integrate with secure payment gateways for transactions.

4. Non-Functional Requirements

i) Performance:

- The platform should load quickly and respond to user actions without significant delays.
- Support concurrent user sessions.

ii) Security:

- Implement secure authentication and authorization mechanisms.
- Protect user data and payment information.

iii) Usability:

- Maintain a user-friendly and intuitive interface.
- Ensure accessibility for users with disabilities.

iv) Scalability:

- The system should scale to accommodate a growing user base and content library.

v) Availability

- Aim for high availability and minimal downtime.

vi) Compatibility

- Ensure compatibility with various web browsers and mobile devices.

vii) Compliance

- Comply with relevant data protection regulations (e.g.,GDPR).
- Adhere to industry standards for e-learning platforms.

viii) Data Privacy

- Safeguard user data and provide data privacy controls.

ix) Error Handling

- Implement robust error handling and logging mechanisms.

x) Accessibility

- Ensure that the platform is accessible to users with disabilities.

5. Goals and Objectives

5.1. Primary Goals

The primary goals of the Online Learning Management System are:

- Provide a personalized learning experience for users.
- Facilitate efficient content delivery and engagement.
- Foster a community of learners and instructors.
- Ensure data security and privacy.

5.2. Secondary Goals

The secondary goals include:

- Monetize the platform through subscriptions and course sales.
- Continuously improve content recommendations.

DESIGN

I. Software Architecture

i. DFD (Data Flow Diagram)

- **Level 0 DFD**

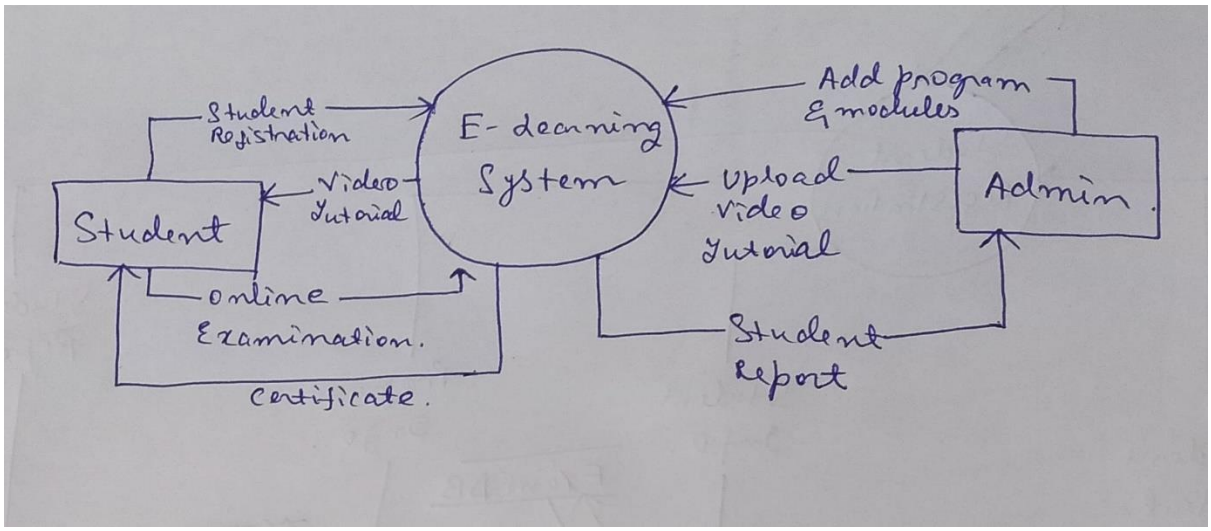


Fig: Level 0 DFD Diagram

At the highest level, we have external entities interacting with the online learning platform. These entities include students, instructors, and administrators. The main processes are also shown.

i) External Entities:

- Student
- Instructor
- Administrator

ii) Processes:

- Course Management
- User Management
- Content Recommendation

- Assessment and Feedback
- Reporting

iii) Data Stores:

- User Database
- Course Database
- Content Repository
- Assessment Database

iv) Data Flows:

- Students request courses and content from the platform.
- Instructors upload content and assessments.
- Administrators manage user accounts.
- The system recommends personalized content to students.
- Assessment results and feedback flow back to the platform.]

- **LEVEL 1 DFD**

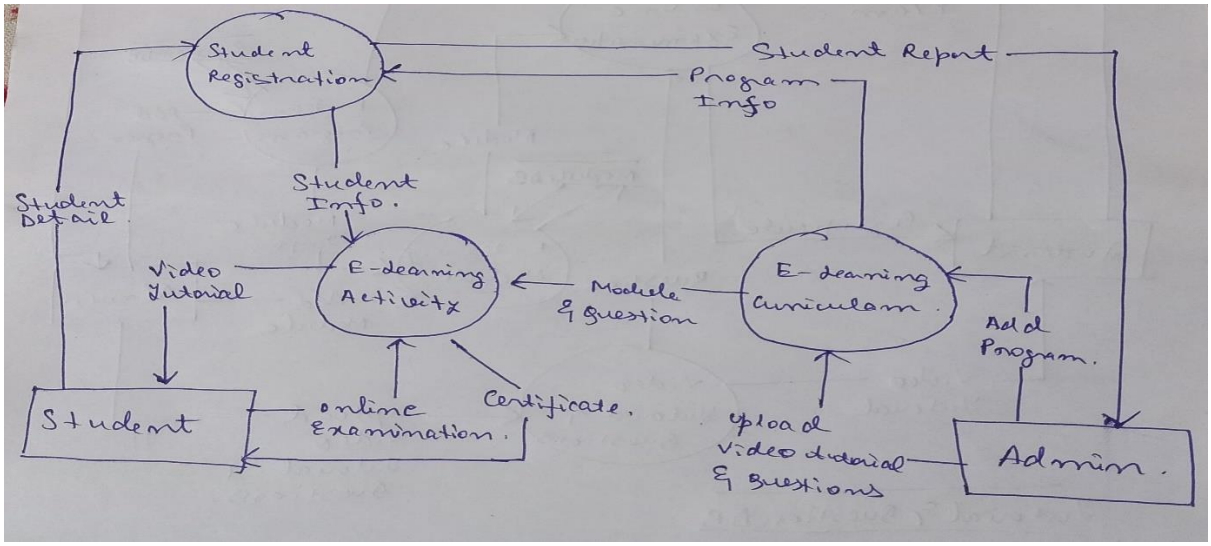


Fig: Level 1 DFD Diagram

let's break some of the main processes into more detailed processes:

i) Course Management:

- Create Course
- Update Course
- Delete Course
- Enroll Student in Course

ii) User Management:

- Register User
- Update User Profile
- Delete User
- Authenticate User

iii) Content Recommendation:

- Analyse User Behaviour
- Generate Personalized Content Recommendations
- Assessment and Feedback:
- Take Assessment

- **LEVEL 2 DFD**

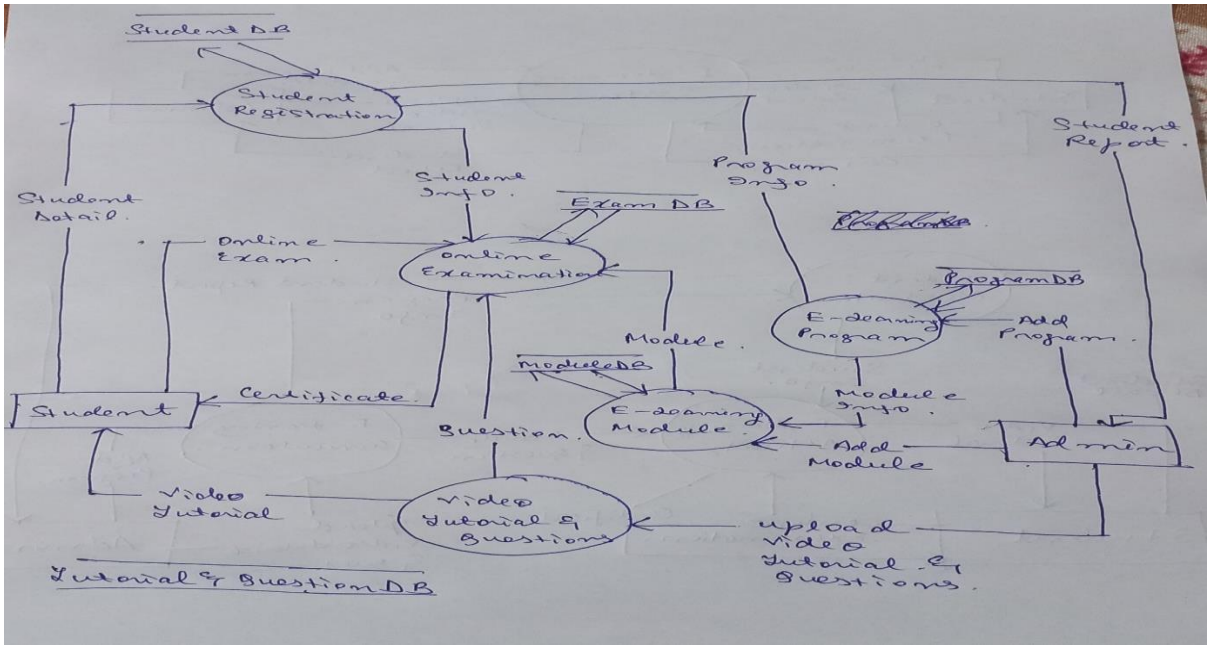


Fig: Level 2 DFD Diagram

Further detailed processes can be broken down for each of the Level 1 processes. For example, under "Create Course" in Course Management, we might have sub-processes like:

- Define Course Details
- Add Content to Course
- Set Assessment Criteria

Similarly, other Level 1 processes can be broken down into more specific sub-processes to provide a comprehensive view of how data flows within the system.

i) External Entities:

- Students interact with the system to access courses and assessments.
- Instructors upload content and assessments.
- Administrators manage user accounts.

ii) Data Stores:

- User Database stores user profiles, authentication information, and progress.
- Course Database stores information about available courses.
- Content Repository contains all the learning materials.
- Assessment Database stores assessments, their results, and feedback.

iii) Data Flows:

- Students request courses, which involve data retrieval from the Course Database and Content Repository.
- Instructors upload content, which involves updating the Content Repository.
- Administrators manage user accounts by interacting with the User Database.
- The system analyses user behaviour using data from the User Database and

Content Repository to generate personalized content recommendations.

- Assessment results and feedback are stored in the Assessment Database.

II) STRUCTURED CHART

i) Project Overview

- Goals and Objectives

- Stakeholders
- Timeline
- Budget

ii). Architecture and Infrastructure

- Cloud-based Infrastructure
- Data Storage and Management
- Security Measures
- API Integration

iii). User Interface and Experience

- User Dashboard
- Content Access
- User Profiles

iv). Content Management

- Content Repository
- Content Curation
- Content Upload and Update
- Content Tagging and Metadata

v). Learning Modules

- Lesson Creation
- Assessments and Quizzes
- Progress Tracking
- Resource Library

vi). User Engagement

- Notification System
- Discussion Forums
- Chatbot Support
- Social Integration

vii). Analytics and Reporting

- User Progress Reports
- Content Popularity Analysis
- User Feedback Analysis
- Learning Effectiveness Metrics

viii). Administrative Tools

- User Management
- Content Moderation
- Access Control
- Data Backup and Recovery

ix). Integration with External Systems

- Learning Management Systems (LMS)
- CRM Integration
- Payment Gateways
- Learning Analytics Tools

x). Testing and Quality Assurance

- Unit Testing
- Integration Testing
- User Acceptance Testing

- Performance Testing

xi). Deployment and Scaling

- DevOps Pipeline
- Scalability Measures
- Load Balancing
- Continuous Integration/Delivery

xiii). User Training and Support

- Training Materials
- Helpdesk and Support Channels
- FAQ and Knowledge Base
- Bug Reporting System

xiv). Security and Privacy

- Data Encryption
- Compliance with Regulations
- User Data Protection
- Security Audits

xv). Project Documentation

- User Manuals
- API Documentation
- Technical Documentation
- Change Log and Versioning

xvi). Project Monitoring and Management

- Progress Tracking
- Issue Management
- Agile Development
- Project Metrics

xvii). User Feedback and Iteration

- User Surveys
- Feedback Analysis
- Continuous Improvement
- Feature Enhancement

xviii). Project Evaluation and Reporting

- Key Performance Indicators (KPIs)
- Project Report
- Lessons Learned
- Future Recommendations

xix). Conclusion and Next Steps

- Project Closure
- Post-Launch Support
- Future Development

III) DATA DICTIONARY

i) Users Table

- user_id: Unique identifier for each user.
- username: User's username for login.
- password: User's hashed and salted password.

- email: User's email address.
- First_name: User's first name.
- last_name: User's last name.
- date_of_birth: User's date of birth.
- profile_picture: URL to the user's profile picture.
- registration_date: Date when the user registered.
- user_type: User's role (e.g., student, teacher, admin).

ii) Courses Table

- course_id: Unique identifier for each course.
- title: Title of the course.
- description: Description of the course.
- instructor_id: User ID of the instructor.
- creation_date: Date when the course was created.
- category: Category of the course (e.g., mathematics, science).
- rating: Average rating of the course.
- price: Course price (if applicable).

iii) Enrollments Table

- enrollment_id: Unique identifier for each enrollment.
- user_id: User ID of the enrolled student.
- course_id: Course ID of the enrolled course.
- enrollment_date: Date when the user enrolled in the course.

iv) Modules Table

- module_id: Unique identifier for each module.
- course_id: Course ID to which the module belongs.
- title: Title of the module.

- description: Description of the module.
- order: Order of the module within the course.

v) Lessons Table

- lesson_id: Unique identifier for each lesson.
- module_id: Module ID to which the lesson belongs.
- title: Title of the lesson.
- content: Lesson content (e.g., text, video URL).
- duration: Duration of the lesson in minutes.

vi) Assessments Table

- assessment_id: Unique identifier for each assessment.
- lesson_id: Lesson ID associated with the assessment.
- type: Type of assessment (e.g., quiz, assignment).
- question_count: Number of questions in the assessment.

vii) User Progress Table

- progress_id: Unique identifier for each user's progress entry.
- user_id: User ID of the student.
- course_id: Course ID for the user's progress.
- module_id: Module ID the user is currently studying.
- lesson_id: Lesson ID the user is currently studying.
- completion_status: Completion status of the lesson/module.

viii) Recommendations Table

- recommendation_id: Unique identifier for each recommendation.
- user_id: User ID for whom the recommendation is generated.
- recommended_course_id: ID of the recommended course.

- recommended_module_id: ID of the recommended module.
- recommended_lesson_id: ID of the recommended lesson.
- timestamp: Date and time when the recommendation was generated.

ix) User Feedback Table

- feedback_id: Unique identifier for each feedback entry.
- user_id: User ID providing feedback.
- course_id: Course ID related to the feedback.
- rating: User's rating for the course.
- comment: User's comments and feedback.

x) System Logs Table

- log_id: Unique identifier for each system log entry.
- user_id: User ID associated with the log entry (if applicable).
- event_type: Type of event (e.g., login, content update).
- timestamp: Date and time of the event.
- description: Description of the event or log entry

USE CASE DIAGRAM

Actors:

- **Admin**: Manages courses, users, and overall system configurations.

- **Instructor**: Creates and manages course content, assignments, and assessments.

- **Student**: Accesses courses, views content, submits assignments, and takes assessments

Use Cases:

Admin Functions:

- **Manage Courses**: Add, delete, update courses in the system.
- **Manage Users**: Add, remove, or modify user accounts (students, instructors, admins).
- **Configure System**: Adjust system settings, permissions, and configurations.

Instructor Functions:

- **Create Course**: Develop new courses, including content, assignments, and assessments.
- **Manage Content**: Upload lectures, documents, videos, etc., for the course.
- **Grade Assignments**: Assess and grade assignments submitted by students.

Student Functions:

- **Enroll in Course**: Join available courses offered by instructors.
- **Access Content**: View lectures, download materials, and participate in discussions.
- **Submit Assignments**: Upload completed assignments for grading.
- **Take Assessments**: Participate in quizzes, exams, or other evaluations.

Relationships:

- **Admin-User Interaction**: The admin manages user accounts, including instructors and students.
- **Instructor-Student Interaction**: Instructors create courses that students can enroll in.
- **Instructor-Admin Interaction**: Admins can assist instructors with technical issues or course-related queries.
- **Student-Course Interaction**: Students access courses and interact with course content.

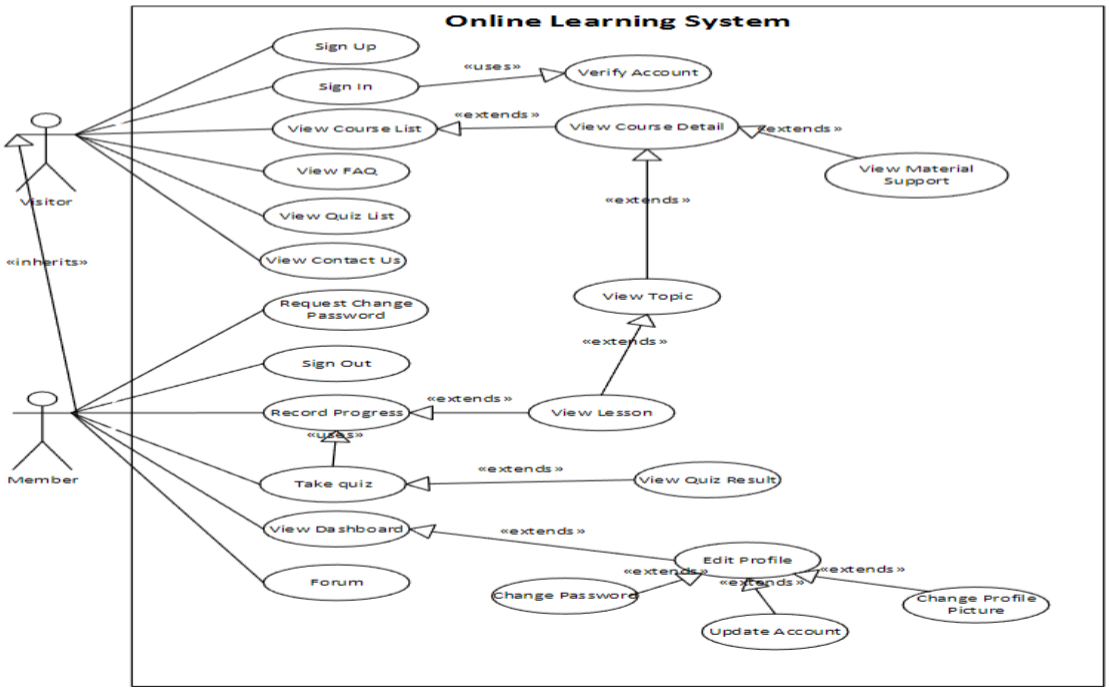


Fig: Use Case Diagram for user of Online Learning System

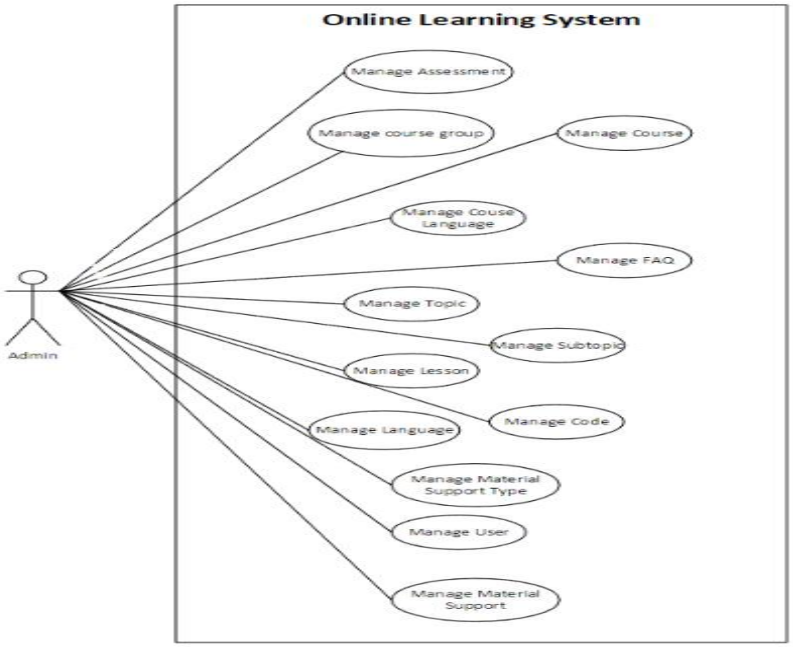


Fig: Use Case Diagram for Admin of Online Learning System

IMPLEMENTATION DETAILS

i) Hardware Used

- Processor: AMD Ryzen 5
- RAM: 16 GB
- Storage: 512 GB

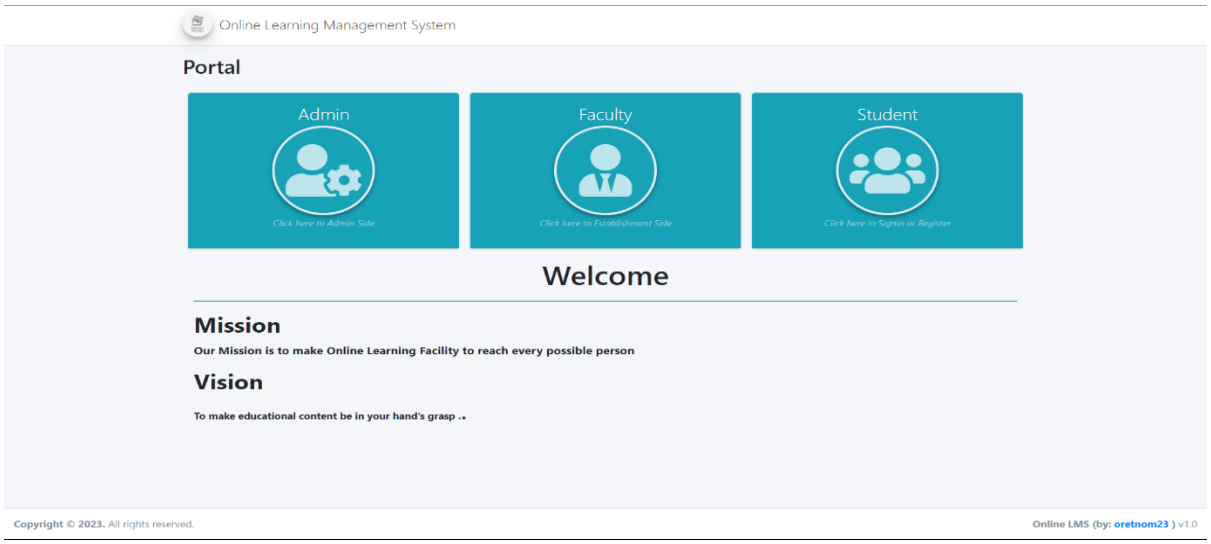
ii) Software Used

- Database: SQL
- Operating System: Windows 11
- Design and Coding Details: PHP
- Documentation Details: MS Word
- Browser: Brave
- Server: XAMPP

Progress in Testing

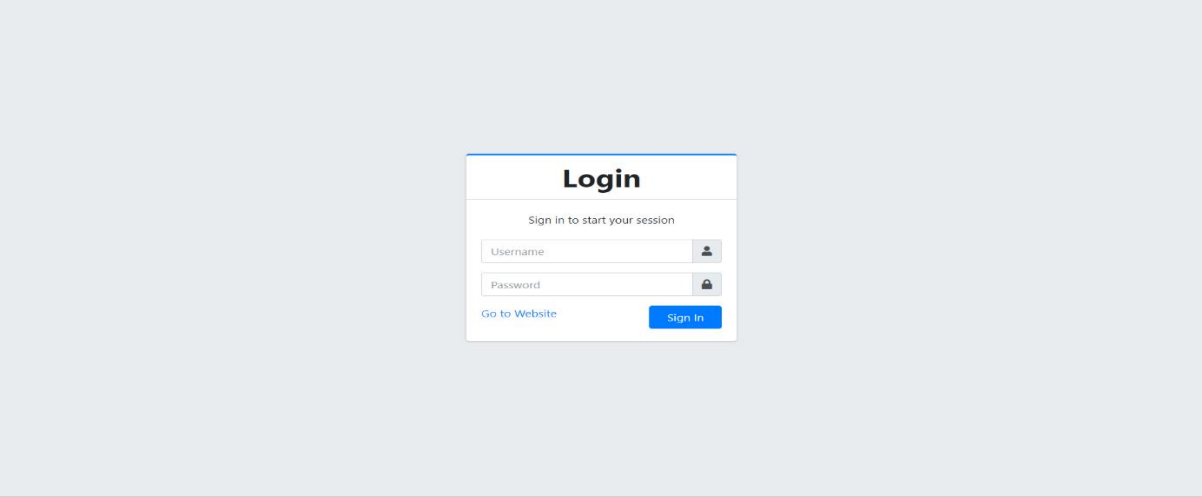
Till now I have completed the following basic testing and I am still working on more features which will take some more but I can assure it will be worth the wait.

- Home Page



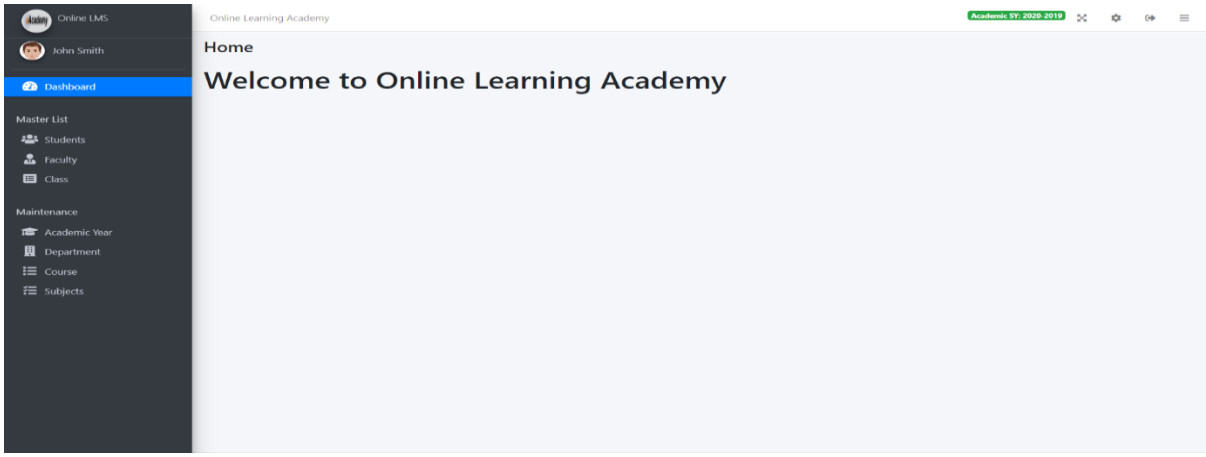
When users enter the website the site purpose is made clear with short description of purpose of site and smooth navigation to the other pages is also provided. In home page you can see three types of login types “Admin, Faculty and Student”

- **Login Page**



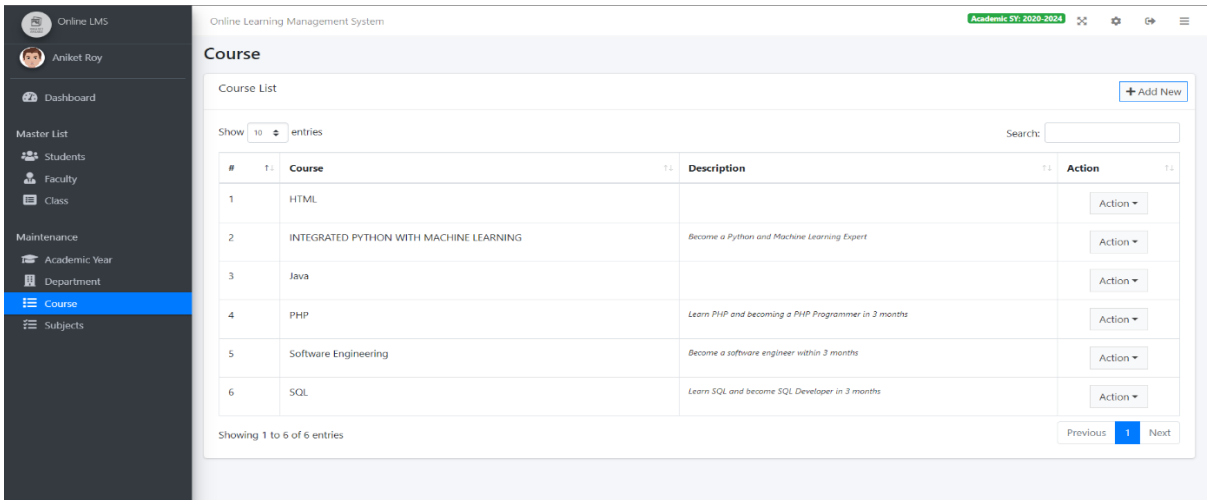
All the registered user’s user their user name and password to login into the system. If they enter wrong credentials error message is displayed else they are redirected to their learning system. When users click on Login in Homepage appropriate information where user need to enter user name and password are displayed without user searching for it.

- **Dashboard**



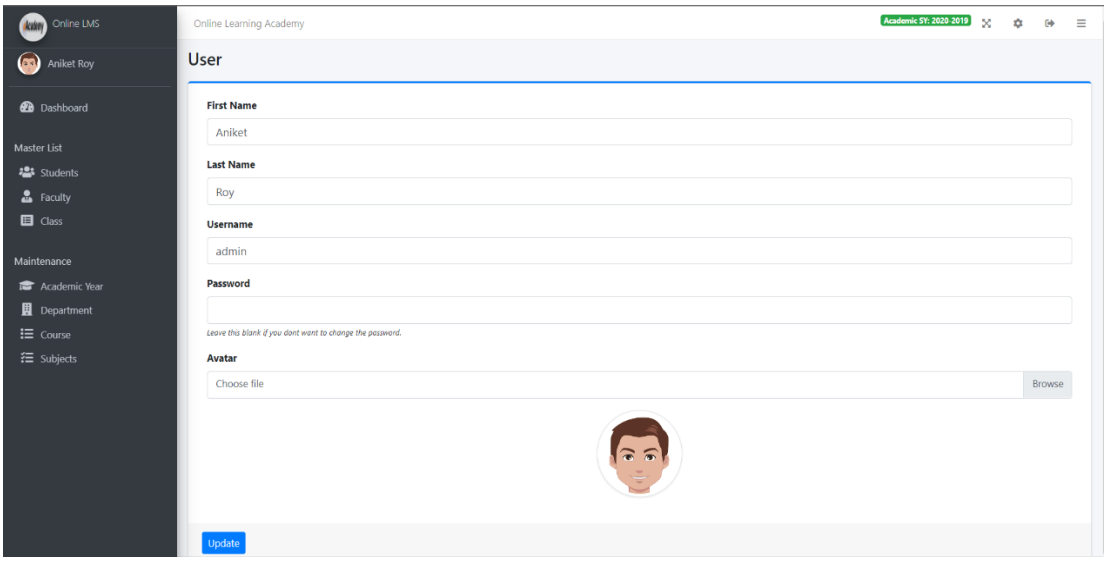
After successful login user is redirected to important information page where user can perform different actions like enrolling, scheduling.

- **Enrolling into Courses**



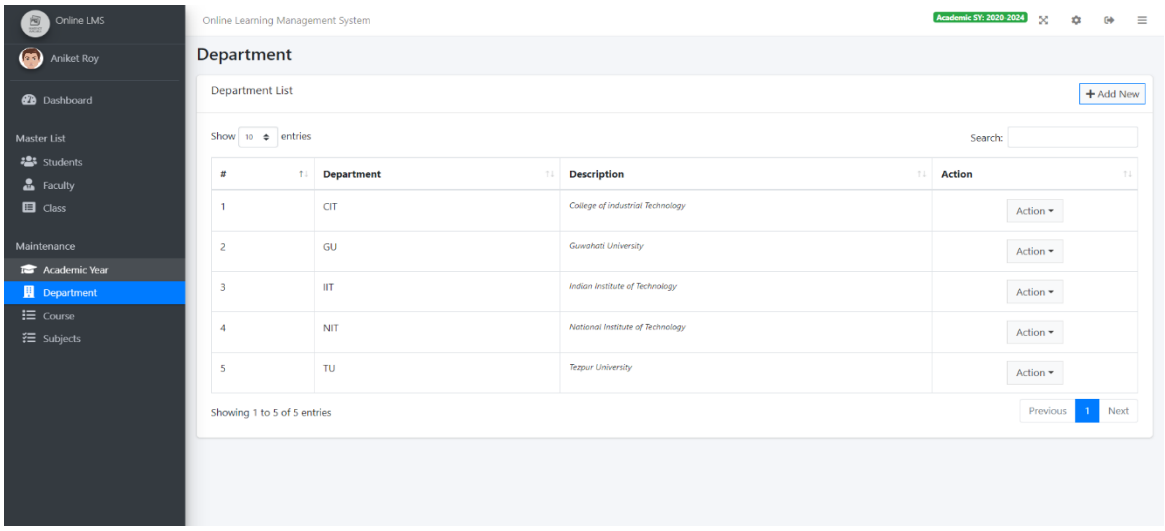
While user is enrolling into courses dropdown lists are provided with appropriate descriptions so that the user is not confused. Also, once user enrolls successfully, confirmation message is displayed.

- **Admin Page**



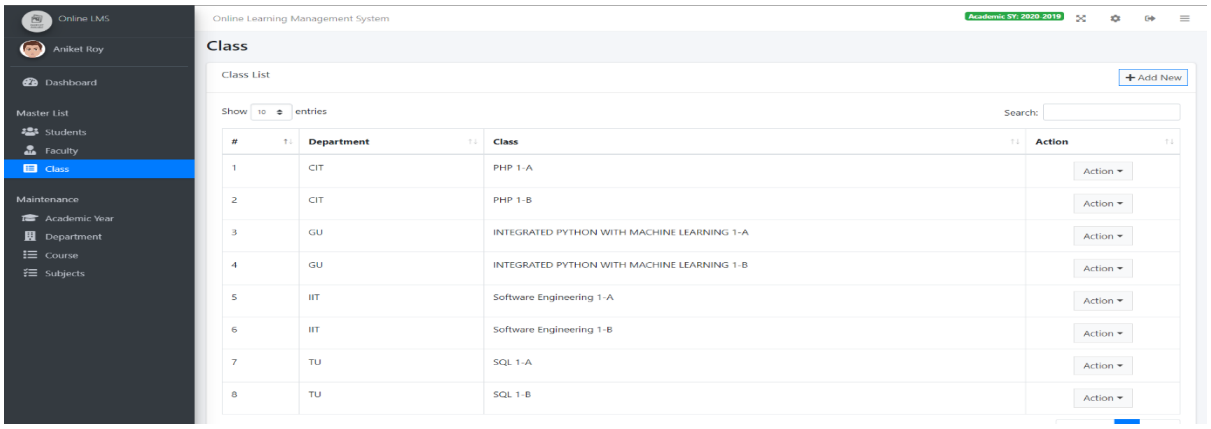
After successful admin login, admin is redirected to admin homepage where scores are updated, events are posted, files are uploaded displaying only required functionality. When admin is uploading the file if file is chosen is displayed, after uploading confirmation message is displayed for admin and smooth navigation to other functionalities are also provided

- **Department List**



This list shows of department list (College Names) which are giving the courses which are listed above. Students who are taking a particular course can see which department/college is teaching the course online.

- **Courses Classes**



The list shows Course Classes where the students are allotted. Students who are allotted in a particular class should join that class

CONCLUSION

The Online Learning System allows unregistered users to register, chat with the admin, read reviews, and view materials. Registered users can enroll in different courses and levels, delete courses, schedule tests, view performance graphs, and contact the admin via email.

In Future I will add features like Admins can update test scores daily, create events related to the training system on the news board, and upload updated materials to the courses. Unregistered users can register for the system, chat with the admin, read reviews, and view materials of low levels of all technologies. They can also schedule tests, view performance graphs, and contact the admin via email for any questions.

Overall, my Online Learning System software will provide a comprehensive and user-friendly platform for learning and development in Future.

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